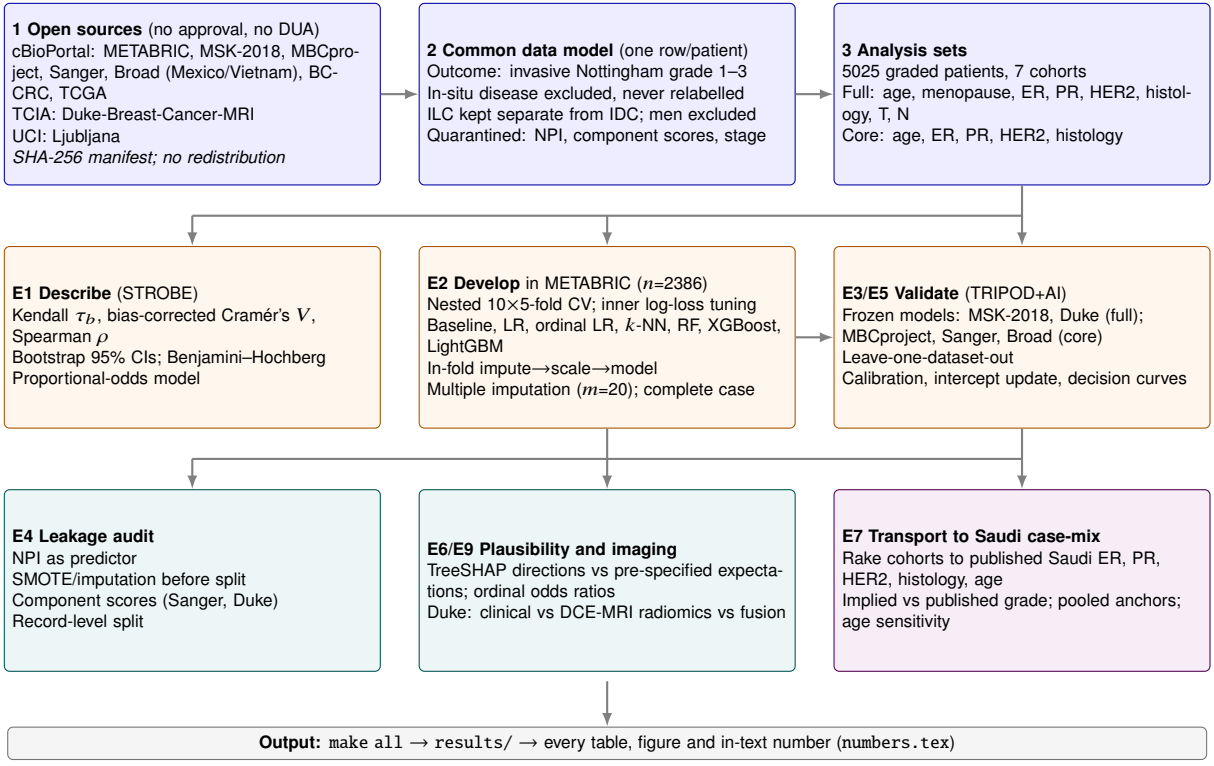
Data
1 Open sources (no approval, no DUA)
cBioPortal: METABRIC, MSK-2018, MBCproject, Sanger, Broad (Mexico/Vietnam), BCCRC, TCGA
TCIA: Duke-Breast-Cancer-MRI
UCI: Ljubljana
SHA-256 manifest; no redistribution
2 Common data model (one row/patient)
Outcome: invasive Nottingham grade 1–3
In-situ disease excluded, never relabelled
ILC kept separate from IDC; men excluded
Quarantined: NPI, component scores, stage
3 Analysis sets
5025 graded patients, 7 cohorts
Full: age, menopause, ER, PR, HER2, histology, T, N
Core: age, ER, PR, HER2, histology
Analysis
E1 Describe (STROBE)
Kendall τ_b, bias-corrected Cramér's V, Spearman ρ
Bootstrap 95% CIs; Benjamini–Hochberg
Proportional-odds model
E2 Develop in METABRIC (n=2386)
Nested 10×5-fold CV; inner log-loss tuning
Baseline, LR, ordinal LR, k-NN, RF, XGBoost, LightGBM
In-fold impute→scale→model
Multiple imputation (m=20); complete case
E3/E5 Validate (TRIPOD+AI)
Frozen models: MSK-2018, Duke (full); MBCproject, Sanger, Broad (core)
Leave-one-dataset-out
Calibration, intercept update, decision curves
Audits & transport
E4 Leakage audit
NPI as predictor
SMOTE/imputation before split
Component scores (Sanger, Duke)
Record-level split
E6/E9 Plausibility and imaging
TreeSHAP directions vs pre-specified expectations; ordinal odds ratios
Duke: clinical vs DCE-MRI radiomics vs fusion
E7 Transport to Saudi case-mix
Rake cohorts to published Saudi ER, PR, HER2, histology, age
Implied vs published grade; pooled anchors; age sensitivity
Output: make all → results/ → every table, figure and in-text number (numbers.tex)